

On the Optimal Design of GDP-Linked Sovereign Bonds with Trend Growth^{*}

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July 2026

Preliminary: please do not circulate

Abstract

JEL Classification

Keywords

^{*}The views expressed herein are those of the authors and should not be attributed to the IMF, its Executive Board, or its management.

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Outstanding questions / roadmap

1. Verify normalization and stationary casting of all agents' problems ✓
2. Correct parametrization of income process ✓
3. Calibration of model with noncontingent debt (this problem does not have ξ as a state variable) ✓
4. Representation of Argentina 2005 GDP warrant. Questions:
 - What to do with the 3% growth threshold? Should we put it at 0 given that we are effectively removing the deterministic trend to have $\mathbb{E}[\exp g] = 1$?
 - Can the 'exceed the baseline path' part be represented with ξ as well?
5. Paper core: moments, prices, and welfare with Arg bond; optimal bond design
6. Two-bond version?

INTRODUCTION

2. MODEL

Endowment There is a single tradable good. The economy receives a stochastic endowment stream of this good Y_t in period t . While the true stochastic process is unknown, all agents in the economy share a *baseline* or approximating model according to which

$$\begin{aligned} Y_t &= \exp(z_t)\Gamma_t \\ \log \Gamma_t &= \log \Gamma_{t-1} + g_t \\ g_t &= \rho_g g_{t-1} + (1 - \rho_g)\mu_g + \sigma_g \epsilon_t^g \\ z_t &= \rho_z z_{t-1} + \sigma_z \epsilon_t^z \end{aligned} \tag{1}$$

where the ϵ 's are *iid* standard Normal random variables.

Estimation Equation (1) can be estimated by maximum likelihood using state-space methods given data on real output Y_t . By taking first differences and calling $\tilde{g}_t = g_t - \mu_g$, we obtain

$$\log Y_t - \log Y_{t-1} - \mu_g = z_t - z_{t-1} + \tilde{g}_t = F S_t \tag{Obs}$$

where $S_t = [z_t, z_{t-1}, \tilde{g}_t]$, $F = [1, -1, 1]$, and

$$\underbrace{\begin{pmatrix} z_t \\ z_{t-1} \\ \tilde{g}_t \end{pmatrix}}_{S_t} = \underbrace{\begin{bmatrix} \rho_z & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & \rho_g \end{bmatrix}}_A \underbrace{\begin{pmatrix} z_{t-1} \\ z_{t-2} \\ \tilde{g}_{t-1} \end{pmatrix}}_{S_{t-1}} + \underbrace{\begin{bmatrix} \sigma_z & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_g \end{bmatrix}}_B \epsilon_t \quad (\text{Evol})$$

In this form, given parameters $(\rho_z, \rho_g, \mu_g, \sigma_z, \sigma_g)$, applying the Kalman filter to the system (Obs)-(Evol) yields the likelihood function. We estimate this state-space model on annual data for Argentina's real GDP between 1900 and 2001. Table 1 summarizes the estimation.

TABLE 1: ESTIMATED INCOME PROCESS

	Parameter	Value
Autocorr. cyclical	ρ_z	0.9369
Std. dev. cyclical	σ_z	0.0222
Autocorr. trend	ρ_g	0.7418
Std. dev. trend	σ_g	0.01807
Mean trend growth	μ_g	0.0067

Note: Parameters reported for a quarterly frequency

State-contingent bonds To represent bonds indexed to the level of income, we define a *cumulative growth rate*, for a gain parameter $1 - \rho$ where $\rho \in [0, 1]$

$$\xi_0 = 1 \quad \text{and} \quad \xi_{t+1} = \rho \xi_t + (1 - \rho) \log \frac{Y_{t+1}}{Y_t} \quad (2)$$

Notice that if $1 - \rho = 1/t$ (in particular, if ρ is not constant), then $\xi_t = \frac{1}{t} \log(Y_t/Y_0)$, the average growth rate so far. Setting $\rho = 0$ makes ξ_t equal the current growth rate, while $\rho \rightarrow 1$ on an infinite history makes $\xi_t \rightarrow \mu_g$, the unconditional average growth rate.

To cast the law of motion of ξ , notice that $\frac{Y_{t+1}}{Y_t} = \frac{\exp(z_{t+1})\Gamma_{t+1}}{\exp(z_t)\Gamma_t} = \exp(z_{t+1} - z_t + g_{t+1})$, and therefore $\xi_{t+1} = \rho \xi_t + (1 - \rho)(z_{t+1} - z_t + g_{t+1})$.

In what follows we present a model in which the debt is potentially indexed to z , g , and ξ . Different designs of state-contingent bonds could mean indexing to only a subset of these variables. Note that ξ is not a state variable of the model unless debt is indexed to it.

Normalization Let lowercase values denote detrended variables, normalized at time t by the lagged trend Γ_{t-1} . For example, $y_t = Y_t/\Gamma_{t-1} = \exp(z_t + g_t)$. Homotheticity of CRRA preferences

implies that $u(C_t) = \Gamma_{t-1}^{1-\gamma} u(c_t)$, implying that a $\Gamma_{t-1}^{1-\gamma}$ term factors out of the government's value functions and making its effective discount factor $\tilde{\beta}(g) = \beta e^{(1-\gamma)g}$ state-dependent. The detrended problem is stationary in the state variables (b, z, g) and ξ (e.g. if debt is indexed to cumulative growth), where $b_t = B_t/\Gamma_{t-1}$ is debt normalized by lagged trend.

Preferences The government makes choices sequentially to maximize the representative household's welfare. At the beginning of a period, assuming it is not already excluded from international capital markets, the government decides whether to repay the debt or default after receiving preference shocks (ϵ_R, ϵ_D) .¹

$$\mathcal{V}(b, z, g, \xi) = \mathbb{E} \left[\max \left\{ v_R(b, z, g, \xi) + \epsilon_R, v_D(b, z, g, \xi) + \epsilon_D \right\} \right] \quad (3)$$

where v_R and v_D respectively denote the value attained after repayment and after default, described below. As is well-known (Chatterjee et al., 2018; Dvorkin et al., 2021), the following closed-forms for the value function and the ex-post probability of default obtain if the ϵ 's follow a Type 1 Extreme Value distribution.

$$\begin{aligned} \mathcal{V}(b, z, g, \xi) &= \chi \log \left(\exp \left(\frac{1}{\chi} v_R(b, z, g, \xi) \right) + \exp \left(\frac{1}{\chi} v_D(b, z, g, \xi) \right) \right) \\ \mathcal{P}(b, z, g, \xi) &= \frac{\exp \left(\frac{1}{\chi} v_D(b, z, g, \xi) \right)}{\exp \left(\frac{1}{\chi} v_R(b, z, g, \xi) \right) + \exp \left(\frac{1}{\chi} v_D(b, z, g, \xi) \right)} \end{aligned} \quad (4)$$

In repayment, the government issues new debt at a price q . Debt is long-term (Leland, 1998; Hatchondo and Martinez, 2009; Arellano and Ramanarayanan, 2012) and stipulates a sequence of (possibly state-contingent) coupon payments: a bond pays $\kappa(z, g, \xi)$ in the following period and then becomes a perfect substitute for $(1 - \delta)$ units of freshly-issued bonds. When debt is non-contingent, we set $\kappa(z, g, \xi) = \bar{\kappa} = r + \delta$.² We use the convention that the government chooses its debt level normalized to units of the following period, b' , which results in a current value of $e^g b'$ (of which $(1 - \delta)b$ corresponds to rollover of existing debt). For numerical stability, we also introduce preference shocks in the debt choice with a small scale parameter χ_b .

¹In non-stationary form, one would need to write $V = \max\{V_R + \Gamma^{1-\gamma}\epsilon_R, V_D + \Gamma^{1-\gamma}\epsilon_D\}$ to keep the variance/scale parameter of the ϵ 's constant.

²This setting of the coupon rate makes the price of a bond expected to repay with probability 1 equal to 1. In other words, it makes the face value of b bonds equal b . When we introduce indexed bonds later, we will keep versions of this anchor $\bar{\kappa}$.

The value of repayment is (when ξ is present as a state variable)

$$\begin{aligned} v_R(b, z, g, \xi) &= \max_{b' \in \mathcal{B}} u(c) + \tilde{\beta}(g) \mathbb{E}[v(b', z', g') + \epsilon_{b'} \mid z, g] \\ \text{subject to } & c + \kappa(z, g, \xi)b = y(z, g) + q_R(b', z, g, \xi)(e^g b' - (1 - \delta)b) \\ & \xi' = \rho\xi + (1 - \rho)(z' - z + g') \end{aligned} \quad (5)$$

which makes the probability of issuing an amount b' proportional to the exponential of the (scaled) objective function, or

$$\phi_b(b' \mid b, z, g) \propto \exp\left(\frac{1}{\chi b} \left(u(c(b', b, z, g)) + \tilde{\beta}(g) \mathbb{E}[v(b', z', g') \mid z, g]\right)\right).$$

In default, the government is excluded from financial markets and suffers a direct output cost $\phi(y)$. The cost is proportional to output but is assumed to depend on the cyclical component only.³ The government then reenters markets with constant hazard ψ and receives a haircut $\hbar$ on the reentry debt. The value of default is

$$\begin{aligned} v_D(b, z, g, \xi) &= u(y(z, g)(1 - \phi(y(z, g)))) + \tilde{\beta}(g) \mathbb{E} \left[\begin{array}{l} \psi \mathcal{V}((1 - \hbar)be^{-g'}, z', g', \xi') \\ + (1 - \psi)v_D(be^{-g'}, z', g', \xi') \end{array} \mid z, g \right] \\ & \xi' = \rho\xi + (1 - \rho)(z' - z + g') \end{aligned} \quad (6)$$

Following [Aguiar and Amador \(2014\)](#), we set the default costs function $\phi(z, g) = \max\{0, -d_0 + \frac{d_0}{d_1}y(z, g)\}$, written so that defaults do not entail costs when $y < d_0$ and the slope of the cost function at d_0 is d_1 .

Lenders The small open economy borrows risk-neutral lenders who fear model misspecification. Lenders are competitive and discount the future (access funds) at a discount rate $\beta_L = \frac{1}{1+r}$. Given a sequence of consumption $c_L(s)$ in state s , they attain a value

$$v_L(s) = \mathbb{E} \left[c_L(s) + \frac{\beta_L}{-\theta} \log \left(\mathbb{E} [\exp(-\theta v_L(s)) \mid z, g] \right) \right] \quad (7)$$

The outer expectation refers to the government's choice of default in the current period, consumption of lenders is the sum of their endowment plus any payments from or to the government⁴

$$c_L(b', b, z, g, \xi) = \begin{cases} \kappa(z, g, \xi)b - q_R(x, z, g, \xi)(b' - (1 - \delta)b) & \text{in repayment} \\ 0 & \text{in default} \end{cases}$$

³This assumption is implicitly made by all studies which filter the data and consider stationary models.

⁴In default, even though the debt stock has a positive price, no payments are taking place

The following Euler equation ensures that lenders hold the bonds

$$\begin{aligned}
q_R(x, z, g, \xi) &= \beta_L \mathbb{E} \left[\begin{aligned} &(1 - \mathbb{1}_{\mathcal{D}'})M_R(x, z', g', \xi') (\kappa(z', g', \xi') + (1 - \delta)q_R(xe^{-g'}, z', g', \xi')) \\ &+ \mathbb{1}_{\mathcal{D}'}M_D(x, z', g', \xi')q_D(xe^{-g'}, z', g', \xi') \end{aligned} \mid z, g \right] \\
q_D(x, z, g, \xi) &= \beta_L \mathbb{E} \left[\begin{aligned} &\psi(1 - \mathbb{1}_{\mathcal{D}'})M_R((1 - \hbar)x, z', g', \xi')(1 - \hbar) (\kappa(z', g', \xi') + (1 - \delta)(q_R((1 - \hbar)x'e^{-g'}, z', g', \xi'))) \\ &+ \psi \mathbb{1}_{\mathcal{D}'}M_D((1 - \hbar)x, z', g', \xi')(1 - \hbar)q_D((1 - \hbar)x'e^{-g'}, z', g', \xi') \\ &+ (1 - \psi)M_D(x, z', g', \xi')q_D(xe^{-g'}, z', g', \xi') \end{aligned} \mid z, g \right]
\end{aligned} \tag{8}$$

where the stochastic discount factor M is given by

$$M_i(x, z', g', \xi') = \frac{\exp(-\theta e^{g'} v_L(xe^{-g'}, z', g', \xi', D = i))}{\mathbb{E} [\exp(-\theta e^{g'} v_L(xe^{-g'}, z', g', \xi', D')) \mid z, g]} \tag{9}$$

3. QUANTITATIVE RESULTS

We parametrize our model at a quarterly frequency following the literature on sovereign debt and default. We rely on external calibration as much as possible. Parameters related to the income process are set according to our estimation described above; and parameters related to risk-free rates, intertemporal substitution, debt maturity, and duration of default spells are set to standard values.

Table 2 describes our parametrization.

We are left with four parameters (the sovereign's discount rate β , the lenders' robustness θ , and the two parameters describing default costs d_0, d_1) to match four moments of the data. We target the average external debt-to-GDP ratio, the average and standard deviation of spreads, and the default frequency. We contrast data for Argentina in 1993Q1:2001Q4 with simulated data from the model. We extract 35-period samples that end just before a default and condition on no default occurring within those 35 periods or two years prior, as is standard in this literature.⁵

The model with non-stationary income and both trend and cycle shocks is able to generate the features of the data that are typically targeted in the sovereign debt literature, which are based on stationary models compared to filtered data. The unconditional frequency of default required to match the spreads in the data, under rational expectations for international creditors, is very large, even larger than that found by stationary models (Chatterjee and Eyigungor, 2012). However, like in stationary models (Pouzo and Presno, 2016), incorporating robustness is enough to match the data along with a much more reasonable default frequency.

⁵Because we include a positive recovery rate, that second conditional is much less relevant.

TABLE 2: BASELINE PARAMETER VALUES

	Parameter	Value
Sovereign's discount factor	β	0.967
Sovereign's risk aversion	γ	2
Preference shock (default) scale	χ	0.0005
Preference shock (issuance) scale	χ_b	0.0005
Lender's robustness	θ	2.885
Risk-free interest rate	r	0.01
Duration of debt	δ	0.05
Income cycle autocorrelation coefficient	ρ_z	0.9369
Standard deviation of z_t	σ_z	0.0222
Income trend autocorrelation coefficient	ρ_g	0.7418
Standard deviation of g_t	σ_g	0.01807
Reentry probability	ψ	0.0385
Default cost: intercept	d_0	0.8564
Default cost: slope	d_1	0.1754

TABLE 3: CALIBRATION TARGETS

	Data	Benchmark	$\theta = 0$	Rational Expectations
Avg. spread (bps)	815	826	1,070	659
Debt-to-GDP (%)	17.4	17.4	20	17.4
Std. dev. spreads (bps)	443	410	840	531
Default freq. (%)	3	3.02	11.7	7.23

Note: Data moments correspond to Argentina 1993Q1:2001Q4. Model moments are computed on pre-default samples, except the default frequency which is calculated on the ergodic distribution. Spread measures are annualized and debt is reported as share of yearly GDP. The column marked 'Benchmark' refers to the full parametrization described in Table 2. The column marked ' $\theta = 0$ ' shuts down robustness starting from that parametrization. The column marked 'Rational Expectations' is a recalibration exercise imposing $\theta = 0$ and dropping the frequency of default as a target moment.

4. DESIGNING STATE-CONTINGENT BONDS

We begin by considering simple designs in which the coupon rate of a bond equals

$$\kappa(x) = \bar{\kappa} \cdot (1 + \alpha x) \mathbb{1}_{x \geq \tau} \quad (10)$$

for x an indexing variable, either the cyclical or the trend component, z or g .

Figure 1 shows the welfare gains from these designs, relative to a plain-vanilla bond which promises constant coupons $\bar{\kappa}$. Welfare gains for indexing on the cycle z are shown on the left panel, while gains for indexing on the trend g are shown on the right panel. In either case, gains can only be achieved for sufficiently low values of τ , meaning that while a threshold for payments can be introduced, it must be sufficiently low.

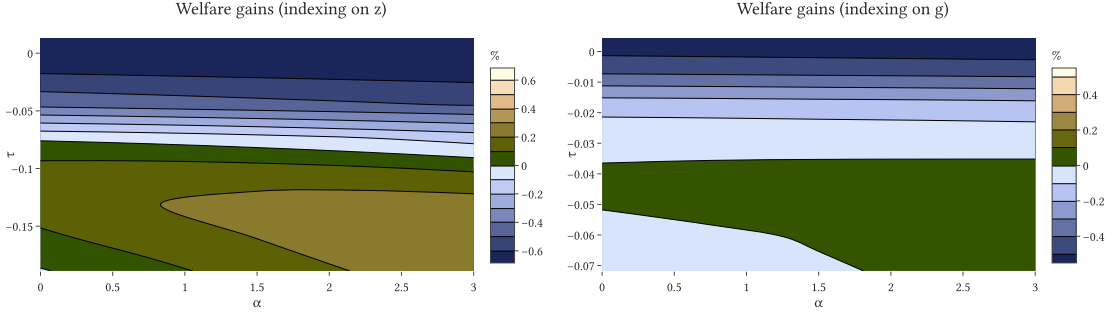


FIGURE 1: WELFARE GAINS FROM INDEXED BONDS

5. VALUE-RECOVERY INSTRUMENTS

We now turn to the design of value-recovery instruments (VRIs). These are a type of state-contingent bond typically used in debt restructuring, often to bridge differences of views between creditors and debtors. These bonds are often designed to share upside risk with creditors in case growth turns out stronger than expected at the time of issuance.

We consider the following design: VRIs are issued only at the time of restructuring, and are indexed using a threshold bond to a “baseline” level of income, which is defined as the expected path of trend income from the time of the restructuring. To make the design concrete, let baseline income (in logs) be $\log Y_t^b = \zeta_t = 0$ at the time of restructuring, growing by the expected trend growth rate μ_g thereafter, so that in normalized log terms $\zeta_{t+1} = \log \frac{Y_{t+1}^b}{\Gamma_t} = \zeta_t + \mu_g - g_t$. The VRI pays a coupon $\bar{\kappa}$ whenever actual income Y_{t+s} exceeds baseline income Y_{t+s}^b , or equivalently,

$$\kappa_W(z, g, \zeta) = \bar{\kappa} \max\{0, y - y^b\} = \bar{\kappa} \max\{0, e^{z+g} - e^{\zeta}\}$$

We assume that the VRIs are only issued at the time of restructuring because the calculation of baseline income depends on the time of issuance.⁶ At the time of restructuring, the government splits up the face value of its outstanding debt into plain-vanilla bonds and VRIs. Our assumptions

⁶Issuing these bonds in normal times would therefore blow up the state space by making bonds issued at different times only imperfect substitutes.

on coupon payments mean that the face value of b units of plain-vanilla bonds is b , and for VRIs we have

$$\text{fv}(w; z, g, \zeta) = wm(z, g, \zeta)$$

where $m(z, g, \zeta)$ is the face value of one unit, which is given recursively⁷ by

$$m(z, g, \zeta) = \bar{\kappa} \max\{0, e^{z+g} - e^{\zeta}\} + \frac{1 - \delta}{1 + r} \mathbb{E}_{z,g}[m(z', g', \zeta')]$$

While the government is in default, it will be useful to keep track of the face value $x = b + \text{fv}(w; z, g, \zeta)$ of all defaulted claims. Upon restructuring, the government splits up this value into a share α of VRIs and $1 - \alpha$ of plain-vanilla bonds. Our previous discussion of face values means that the amounts b_R and w_R issued at the time of restructuring, after a haircut $\hbar$, must satisfy

$$b_R = (1 - \alpha)(1 - \hbar)x; \quad \text{fv}(w_R; z, g, 0) = \alpha(1 - \hbar)x$$

This discussion is summarized by the following Bellman equations for the government's problem in the presence of VRIs. The repayment versus default choice now features the total face value x

$$v(b, w, z, g, \zeta) = \max \left\{ v_R(b, w, z, g, \zeta) + \epsilon_R, v_D(x_D(b, w, z, g, \zeta), z, g) + \epsilon_D \right\} \quad (11)$$

where $x_D(b, w, z, g, \zeta) = b + \text{fv}(w; z, g, \zeta)$ is the face value of defaulted claims. Note that the function $x_D(\cdot)$ links the state in repayment, (b, w, z, g, ζ) , with the state in default, (x, z, g) . As before the probability of default is

$$\mathcal{P}(b, w, z, g, \zeta) = \frac{\exp\{v_D(x_D(b, w, z, g, \zeta), z, g)/\chi\}}{\exp\{v_D(x_D(b, w, z, g, \zeta), z, g)/\chi\} + \exp\{v_R(b, w, z, g, \zeta)/\chi\}}$$

In repayment, the government chooses the issuance of plain-vanilla bonds b' , but no new VRIs are issued and $(1 - \delta)$ units survive after the coupon payment. In normalized terms, $w' = (1 - \delta)e^{-g}w$. The value of repayment is then

$$\begin{aligned} v_R(b, w, z, g, \zeta) &= \max_{b'} u(c) + \tilde{\beta}(g) \mathbb{E} [v(b', w', z', g', \zeta') + \epsilon_{b'} \mid z, g] \\ \text{subject to } c + \kappa b + \kappa_W(z, g, \zeta)w &= y(z, g) + q_R(b', w, z, g, \zeta)(e^g b' - (1 - \delta)b) \\ w' &= (1 - \delta)e^{-g}w \\ \zeta' &= \zeta + \mu_g - g \end{aligned} \quad (12)$$

In default, there are no current payments, but there are important subtleties in timing. The government starts a period of exclusion with a defaulted face value x and income (z, g) . It first receives a reentry shock which restores market access, and arrives with probability ψ . Upon

⁷Note that the Bellman equation for m is a contraction since $(1 - \delta)/(1 + r) < 1$.

reentry, the haircut $\hbar$ is applied to the value x , which is then split into plain-vanilla bonds and VRIs; and the baseline income is reset to $\zeta = 0$. At this point, the government can choose to default again, prolonging the exclusion period. The value of entering a period in exclusion is

$$v_D(x, z, g) = u(y(z, g)(1 - \phi(y(z, g)))) + \tilde{\beta}(g) \mathbb{E} \left[\begin{array}{l} (1 - \psi)v_D(e^{-g'}x, z', g') \\ + \psi v(b_R, w_R, z', g', 0) \end{array} \mid z, g \right] \quad (13)$$

where $b_R = e^{-g'}(1 - \alpha)(1 - \hbar)x$ and $fv(w_R; z', g', 0) = e^{-g'}\alpha(1 - \hbar)x$. Similarly to the $x_D(\cdot)$ function introduced above, the recovered values b_R and w_R (and the reset $\zeta = 0$) link the state in default (x, z, g) with the state in repayment (b, w, z, g, ζ) .

To evaluate the welfare effects of a VRI issuance, we compare the value function of the government in the equilibrium with only noncontingent debt with that in the equilibrium with noncontingent debt and VRIs. We consider three possible criteria: (i) the VRI is first introduced in a restructuring, therefore the comparison in value functions is taken as an average over the ergodic distribution of the equilibrium without VRIs, conditional on a restructuring event, (ii) the VRI is introduced at an arbitrary date, therefore the comparison is taken as an average over the ergodic distribution of the equilibrium without VRIs and comparing with the value of holding no VRIs, $x = 0$, (iii) the VRI is introduced at the start of the economy, so the comparison is taken at $b = x = 0$ and averaging over the ergodic distribution of income (z, g) .

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A. DETAILS ABOUT NORMALIZATION AND ROBUSTNESS

To cast the model in stationary form we normalize by Γ_{t-1} . Let lowercase denote normalized values. For example, $y_t = \exp(z_t + g_t)$. For the value functions, notice that

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma} = \Gamma^{1-\gamma} \frac{(C/\Gamma)^{1-\gamma}}{1-\gamma} = \Gamma^{1-\gamma} u(c)$$

and thus in a typical Bellman equation we can guess and verify forms like $\Gamma_0^{\gamma-1} V(B, z, g, \Gamma_0) = v(B/\Gamma_0, z, g) = v(b, z, g)$, where Γ_0 is the previous period trend so that $\Gamma = \Gamma_0 e^g$:

$$V(B, z, g, \Gamma_0) = \max_{B'} u(C) + \beta \mathbb{E} [V(B', z', g', \Gamma)]$$

$$V(B, z, g, \Gamma_0) = \max_{B'} \Gamma_0^{1-\gamma} u(c) + \beta \mathbb{E} [V(B', z', g', \Gamma)]$$

$$\Gamma_0^{\gamma-1} V(B, z, g, \Gamma_0) = \max_{B'} u(c) + \beta \mathbb{E} [(\Gamma_0/\Gamma)^{\gamma-1} \Gamma^{\gamma-1} V(B', z', g', \Gamma)]$$

$$v(b, z, g) = \max_{b'} u(c) + \beta \mathbb{E} [e^{(1-\gamma)g} v(b', z', g')]$$

while for the budget constraint we have

$$C + \kappa B = Y + q(B' - (1 - \delta)B)$$

$$c + \kappa b = y + q(B'/\Gamma_0 - (1 - \delta)b)$$

$$c + \kappa b = y + q(b'(\Gamma/\Gamma_0) - (1 - \delta)b)$$

$$c + \kappa b = y + q(b' e^g - (1 - \delta)b)$$

where of course the second line uses the definition of $b' = B'/\Gamma$.

Debt prices Consider one lender (with robustness parameter θ/Γ) with current holdings of debt $a = A/\Gamma$ at aggregate state s . To cast the problem in stationary form, once more guess-and-verify a form $V = \Gamma v$

$$V(A, s) = \max_{C, A'} C + \frac{\beta_L}{-\theta/\Gamma} \log (\mathbb{E} [\exp (-(\theta/\Gamma) V(A', s')) \mid s])$$

$$\Gamma v(a, s) = \max_{c, x_a} \Gamma c + \frac{\beta_L}{-\theta/\Gamma} \log (\mathbb{E} [\exp (-(\theta/\Gamma) \Gamma v(e^{-g'} x_a, s')) \mid s])$$

$$v(a, s) = \max_{c, a'} c + \frac{\beta_L}{-\theta} \log (\mathbb{E} [\exp (-\theta e^{g'} v(e^{-g'} a', s')) \mid s])$$

Since $1/\theta$ is the marginal cost of relative entropy for the evil agent, it seems sensible that it would scale with the size of the economy (see [Cagetti et al., 2002](#), for a similar discussion).

To obtain the debt price equation (8), start from the lender's value function

$$v(a, s) = \max_{c, a'} c + \frac{\beta_L}{-\theta} \log \left(\mathbb{E} \left[\exp \left(-\theta e^{g'} v(e^{-g'} a', s') \right) \mid s \right] \right)$$

subject to $qa' + c = R(s)a + w$

The envelope condition is $v'(a, s) = R(s)$ (risk-neutrality removes any references to the derivative of the utility function) and the first-order conditions are

$$\lambda = 1$$

$$\lambda q = \frac{\beta_L}{-\theta} \frac{1}{\mathbb{E} \left[\exp \left(-\theta e^{g'} v(e^{-g'} a', s') \right) \right]} \mathbb{E} \left[-\theta e^{g'} \exp \left(-\theta e^{g'} v(e^{-g'} a', s') \right) e^{-g'} v'(e^{-g'} a', s') \right]$$

or, in the stochastic discount factor notation of equation (9),

$$q = \beta_L \mathbb{E} \left[\frac{\exp \left(-\theta e^{g'} v(e^{-g'} a', s') \right)}{\mathbb{E} \left[\exp \left(-\theta e^{g'} v(e^{-g'} a', s') \right) \right]} R(s') \right] = \mathbb{E} [M(a', s') R(s')]$$